

Project: Causal Machine Learning - Effect of 401(k) Eligibility

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Contributors:

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1. Introduction

This report investigates the causal effect of eligibility for a 401(k) retirement savings plan on accumulated net financial assets. The analysis utilizes data from the Survey of Income and Program Participation, available via the pension dataset in the R `hdm` package (9915 observations). All analyses are based on debiased/targeted machine learning methods, employing 5-fold cross-fitting for nuisance parameter estimation. SuperLearner was used for nuisance estimation, with a full library (`SL.mean`, `SL.glm`, `SL.glmnet`, `SL.randomForest`, `SL.gbm`) for Q4–Q5 and a reduced one (`SL.glmnet`, `SL.gam`) for Q1–Q3 to speed up computation.

2. Question 1: Exchangeability of 401(k) Eligibility

The primary challenge in estimating the causal effect of 401(k) plans is potential confounding due to non-random selection into eligibility and participation. It is argued that eligibility for a 401(k) plan (A) can be considered exchangeable with respect to potential net financial assets (Ya), conditional on a set of pre-eligibility covariates (L), particularly income.

This assumption can be represented as a Directed Acyclic Graph (DAG) in Figure 0.

The diagram illustrates the assumed causal relationships. 'U' represents unobserved factors like savings preference, 'L' represents observed confounders, 'A' is 401(k) eligibility, and 'Y' is net financial assets. The primary causal effect of interest is $A \rightarrow Y$ (red arrow). By conditioning on L (highlighted node), we aim to block the backdoor (non-causal) paths from A to Y (e.g., $A \leftarrow L \leftarrow U \rightarrow Y$ and $A \leftarrow L \rightarrow Y$), making it more plausible to estimate the true effect of A on Y. This assumes no unmeasured confounders U directly affect A after accounting for L.



Question 2: Estimating the ATE/ATT of Eligibility on Net Financial Assets

We started by calculating the observed difference in net financial assets between eligible and ineligible groups. The eligible group had a mean net asset value of \$30,347, while the ineligible group had \$10,788.04 (t-test, $p < 2.2 \times 10^{-16}$). However, this \$20K difference does not reflect a causal effect and likely suffers from **confounding bias**, as indicated by the DAG discussed above.

Using **AIPW**, we estimated the Average Treatment Effect (**ATE**), both manually and using the `npcausal` package. This addresses the question: *If everyone were eligible versus nobody, how much would net assets increase on average?* We obtained an ATE of **\$7,710 ±**

1,181 (95% CI of [\$5397, \$10023]). Note that this is smaller than the naive analysis above, suggesting that a significant portion of the raw difference is likely due to confounding factors; individuals who are eligible for 401(k) plans likely possess other characteristics (e.g., higher income, different savings preferences correlated with covariates) that predispose them to accumulate more assets, irrespective of 401(k) eligibility itself. Another explanation for the difference between the adjusted and unadjusted estimate, is that we have erroneously adjusted for a mediator instead of a confounder. This would block the indirect path from eligibility to assets, reducing the measured effect. Propensity scores ranged from 0.06 to 0.85 in both groups, confirming the positivity assumption and good overlap between the eligible and ineligible groups. Boxplots comparing the inverse probability weights showed that some eligible individuals had moderately outlying weights, most of which could be linked to very low-income cases (e.g., income = \$0) and income was strongly negatively correlated with the weights in the eligible group, indicating that low-income individuals are rarely eligible. One might thus question how realistic the ATE scenario is.

A more meaningful question might be - *among those that were eligible, what was their average financial gain from eligibility?* - as this evaluates the success of the plan. This Average Treatment effect on the Treated (**ATT**) was estimated at **\$10,367 ± 1,345** (95% CI of [\$7730, \$13004]) and suggested a 50% increase in net financial assets compared to a world where these individuals were not eligible for the plan.

A similar reasoning applies to the **TMLE** results: **ATE = \$8,330 ± 1,143**, **ATT = \$10,416 ± 1,746**. The former was estimated using the tmle package, the latter manually using ATT-specific clever covariates and fitting the fluctuation model.

4. Question 3: Testing for Conditional Covariance ($E\{\text{Cov}(A, Y|L)\}=0$)

To assess whether eligibility for a 401(k) plan (A) and net financial assets (Y) remain conditionally associated after adjusting for covariates (L), we tested the null hypothesis that $E\{\text{Cov}(A, Y|L)\}=0$. The efficient influence curve for $\theta=E\{\text{Cov}(A, Y|L)\}$ was used for this estimation.

- The estimated conditional covariance $\theta^{\wedge}$ was **\$1,747 ± 254**
- This resulted in a Z-score of 6.88, with a 2-sided p-value of 6×10^{-12}

Interpretation: The small p-value provides strong evidence to reject the null hypothesis. This provides strong evidence that, even after adjusting for covariates L, there remains a positive association between 401(k) eligibility and net financial assets when averaging across the covariate distribution. Under conditional exchangeability assumption (Q1), this provides evidence consistent with a causal effect of plan eligibility on net financial assets.

5. Question 4: Heterogeneous Effects using R-learner

The R-learner was used to estimate the Conditional Average Treatment Effect (CATE), $\tau(X)=E[Y(1)-Y(0)|X]$, of 401(k) eligibility on net financial assets, as a function of all covariates L.

The distribution of estimated individual CATEs is shown in Figure 1.

The histogram shows that most estimated individual effects of eligibility are positive. The distribution appears to be concentrated slightly above zero for many individuals, with a tail indicating that for some individuals, the predicted effect is larger. There is a very tall, thin bar at the extreme right of the plot, suggesting either a few individuals with very high estimated CATEs or a plotting artifact due to outliers; this warrants cautious interpretation or further investigation of those specific CATE values if refining the analysis.

The variation of these CATEs with age, income, and years of education is visualized in Figures 2a-2c.

- **Age (Figure 2a):** The estimated CATE of 401(k) eligibility generally appears to increase with age from younger working years, peaking in middle age (around 40-50), and then potentially plateauing or slightly declining for older individuals.
- **Income (Figure 2b):** The relationship between CATE and income appears non-linear. The effect seems to increase with income at lower income levels, but this trend may reverse or become more complex at higher income levels, as indicated by the LOESS smoother.
- **Education (Figure 2c):** The estimated CATE tends to increase with years of education, suggesting individuals with more education may experience a larger positive impact on asset accumulation from 401(k) eligibility.

These plots suggest that the effect of 401(k) eligibility is indeed heterogeneous and varies systematically with key demographic and economic characteristics. This suggests that the test for the conditional variance (Q3) should be interpreted with care, especially when it comes to income and education.

6. Question 5: Predicting Counterfactual Y1 and Estimating Prediction Error

A doubly robust (DR-learner) approach was used to predict the counterfactual net financial assets if eligible (Y_1) as a function of age, income, and years of education, while controlling for the full set of covariates L in the construction of the pseudo-outcome.



The distribution of these predicted potential outcomes ($Y_{\sim 1}$) is shown in Figure 3.

The distribution of predicted Y_1 is highly right-skewed. This suggests that for most individuals, the predicted potential assets under eligibility are concentrated at lower values, but there is a long tail representing a smaller number of individuals for whom predicted Y_1 is substantially higher.

We then estimated the counterfactual prediction error, defined as $E\{(Y_1 - Y_{\sim 1})^2\}$.

- The estimated error was **3,061,861,579** (SE: 627,069,756).
- The 95% confidence interval for this error was [\$1,832,804,857, \$4,290,918,300].

The estimated expected squared difference between the counterfactual assets under eligibility (Y_1) and our model's predictions ($Y_{\sim 1}$) is very large. The square root of this value, approximately \$55,330, can be seen as a form of root mean squared error, indicating the typical magnitude of prediction error in dollar terms. This suggests that while the model can

generate predictions, there is considerable uncertainty or deviation in these predictions of individual counterfactual asset levels.

7. Quality Checks

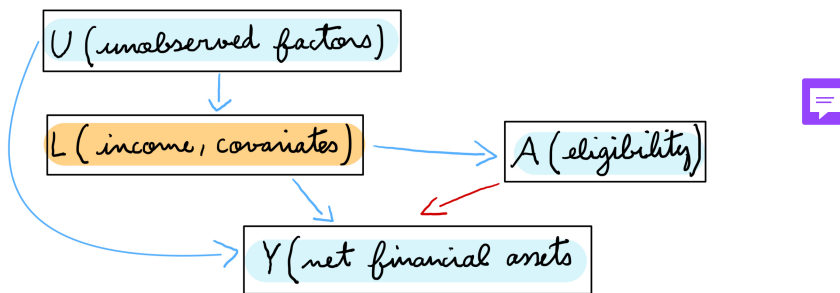
Nuisance parameters (propensity scores and conditional mean outcomes) were estimated using SuperLearner with 5-fold cross-fitting. The distribution of the estimated propensity scores ($P(A=1|L)$) (Figure 4) indicates good overlap and no extreme values very close to 0 or 1, supporting the stability of estimators relying on inverse probability weighting.

8. Conclusion

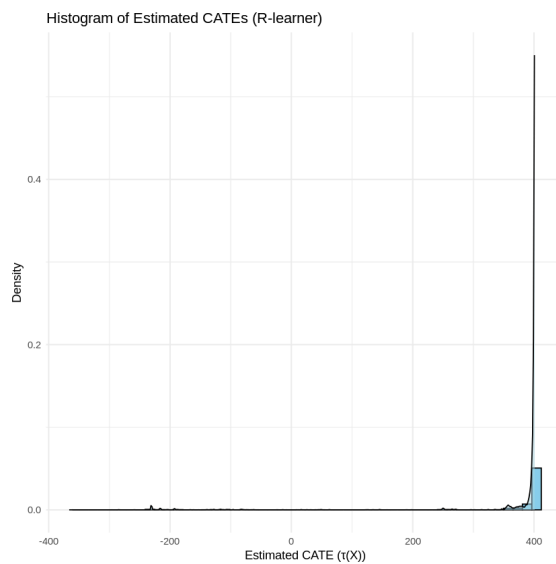
This analysis suggests that 401(k) eligibility has a positive causal effect on net financial assets, estimated to be around \$7,660 on average after controlling for confounders. This effect appears to be heterogeneous, varying with age, but particularly with income, and education. While counterfactual asset levels under eligibility can be predicted, the associated prediction error is substantial.

Figures

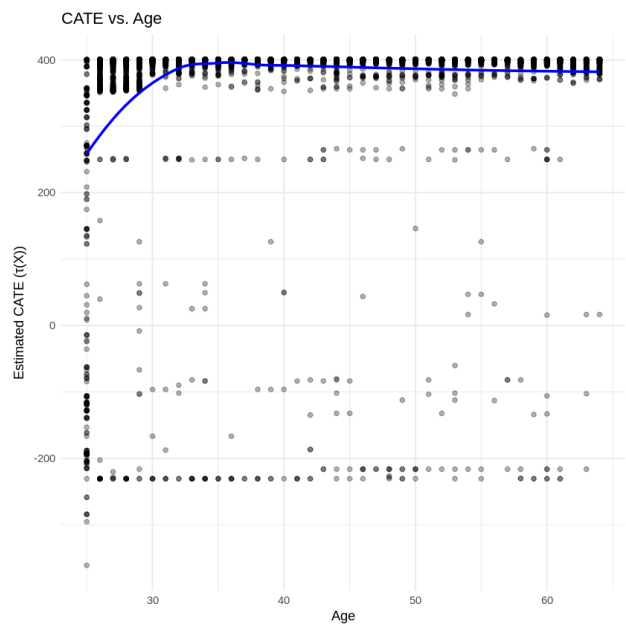
Figure 0: Causal Diagram for 401(k) Eligibility and Assets.



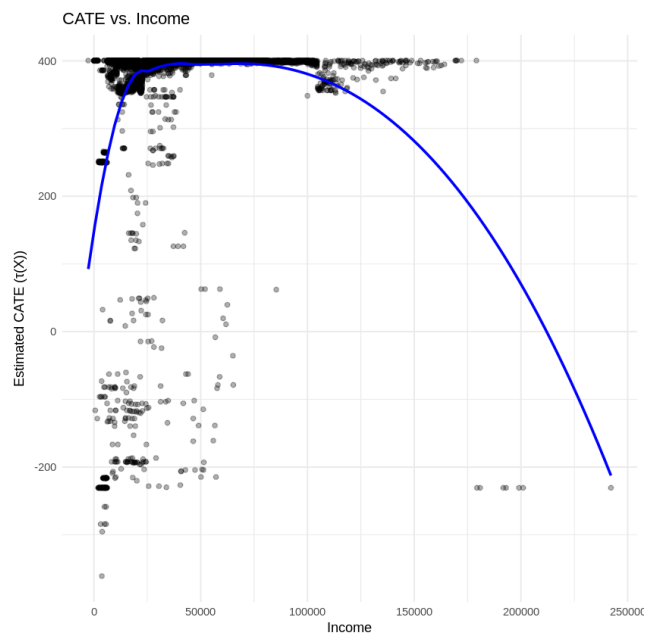
- Figure 1: Histogram of Estimated CATEs ($\tau(X)$) from R-learner.



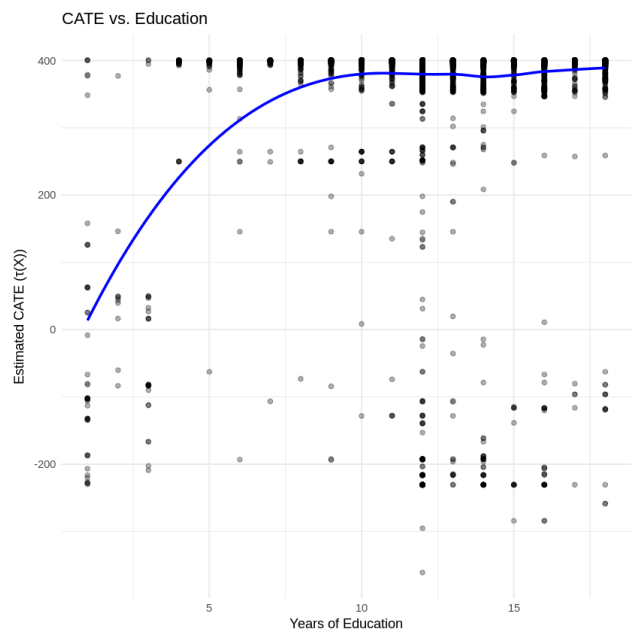
● **Figure 2a: Estimated CATE vs. Age.**



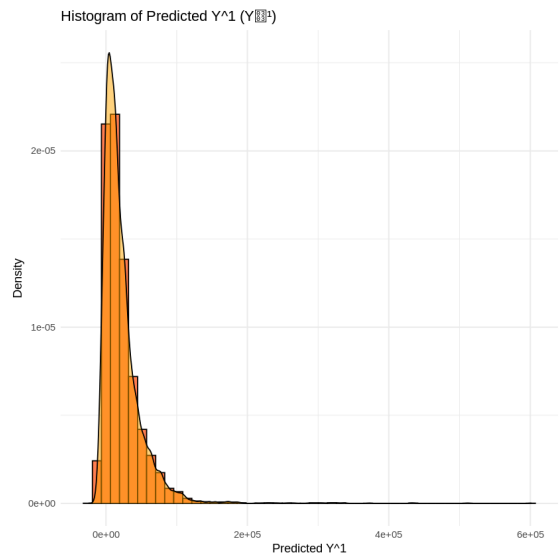
● **Figure 2b: Estimated CATE vs. Income.**



● **Figure 2c: Estimated CATE vs. Years of Education.**



● **Figure 3: Histogram of Predicted Y1 (Net Financial Assets if Eligible).**



● **Figure 4: Distribution of Estimated Propensity Scores $P(A=1|L)$.**

